

PROFESSIONAL SUMMARY

Computer Science graduate (BS CS, De La Salle University–Dasmariñas, 2021–2025) building applied AI workflows, multi-agent systems, and full-stack web applications. Experienced in orchestrating cloud and local LLMs (Groq and Gemini, with an optional local Ollama mode on an AMD RX 6600 XT), developing automated tool pipelines in Python, and building responsive, accessible interfaces with React, TypeScript, and modern CSS.

PROFESSIONAL WORK EXPERIENCE

UI/UX & Frontend Engineer Intern · ROC.PH Digital Marketing Services (roc.ph) · General Trias, Cavite, Philippines

June 2024 – August 2024

- Designed and prototyped responsive web and mobile interfaces in Figma, strictly applying Nielsen Norman Group usability heuristics and design token standards.
- Translated design specifications into accessible, semantic HTML5, modern CSS, Tailwind CSS, and React components deployed across production client sites.
- Partnered within cross-functional agile sprints, participating in daily standups, design reviews, Git version control workflows, and rapid client feedback turnaround loops.

KEY TECHNICAL IMPACT

- Engineered a bilingual voice pipeline: FFmpeg 16 kHz normalization, Groq Whisper transcription with an OpenAI fallback, and Edge-TTS voices that start speaking as each sentence is written (time to first audio 4.60 → 1.41 s p50, N=10).
- Architected 5 asyncio Discord agents with per-agent cloud models and Groq/Gemini fallback, hybrid RAG (Gemini embeddings + BM25, fused with RRF), and Gmail triage across 10 labels.
- Built a zero-dependency lessons-learned memory in the Python standard library with field-weighted BM25 (0.81 ms p50 recall; recall@3 0.90 on 30 held-out queries).
- Developed responsive, accessible web components in React, TypeScript, and modern CSS during a commercial frontend internship at ROC.PH.

FEATURED ENGINEERING PROJECTS

Manas: Ciel

Voice AI Copilot & Real-Time Cognitive HUD · Solo Project

Built a bilingual JA/EN voice copilot with streaming speech, a custom FFmpeg voice filter, live weather (wttr.in), web search (DDGS), Obsidian note tools, and a WebSocket HUD.

LSFM AI HQ

Autonomous Multi-Agent Operations System · Solo Project

Architected 5 Discord agents with Groq + Gemini routing (optional local Ollama mode), a headless-Edge ATS resume builder, and OAuth2 inbox triage.

Patriot

University Community Matchmaking Mobile App · BS Computer Science Capstone

Engineered a cross-platform matchmaking app in React Native/TypeScript with AWS Lambda, DynamoDB and Cognito, Redis caching for session state, and interest-based matching.

Cognitive Memory Core

Lessons-Learned Memory for AI Coding Assistants · Solo Project

Built a lessons-learned memory for AI coding assistants: standard-library BM25 retrieval before each task, with repeated lessons compiled into rules.

EDUCATION

Bachelor of Science in Computer Science (BS CS)

2021 – 2025

De La Salle University–Dasmariñas (DLSU-D)

Distributed Systems, Algorithms & Complexities, Software Engineering, Database Management Systems, Human-Computer Interaction (HCI), Object-Oriented Programming.

TECHNICAL PROFICIENCIES & CAPABILITIES

AI & LLM Orchestration:	Multi-Agent Workflows, Web Search Tools (DDGS), Voice Pipelines (Whisper STT, Edge-TTS, FFmpeg filters), Hybrid RAG (Dense + BM25, RRF), Local LLM Inference (Ollama), Cloud LLM APIs (Groq, Google Gemini, OpenAI Whisper), Tool & Function Calling, Structured JSON Output, BM25 Retrieval
Languages & Backend:	Python 3.11 (asyncio, OOP, pure standard library), TypeScript, JavaScript (ES6+), Node.js, Express, aiohttp, REST APIs, WebSocket, OAuth2, Headless Chromium Automation, Git / GitHub
Frontend & Design:	React.js, React Native, HTML5, Modern CSS3, CSS Custom Properties & Design Tokens, Web Audio API, Canvas 2D, Tailwind CSS, Responsive Web Design, Cross-Browser Compatibility, Figma (Auto-layout, Components, Variants, Design Tokens), Wireframing, Information Architecture, WCAG 2.1 Accessibility Standards, Usability Testing